

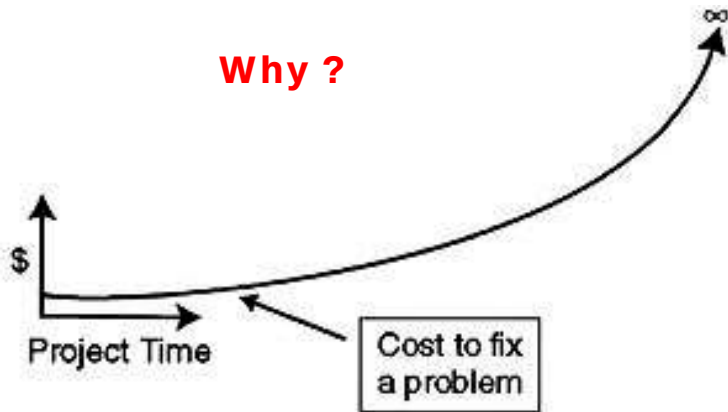
System Driven Hardware Design

- Testing -

International Master of Science

Prof. Dr.-Ing. Stephan Bannwarth

Ensuring a working system during development, assembling and lifetime.



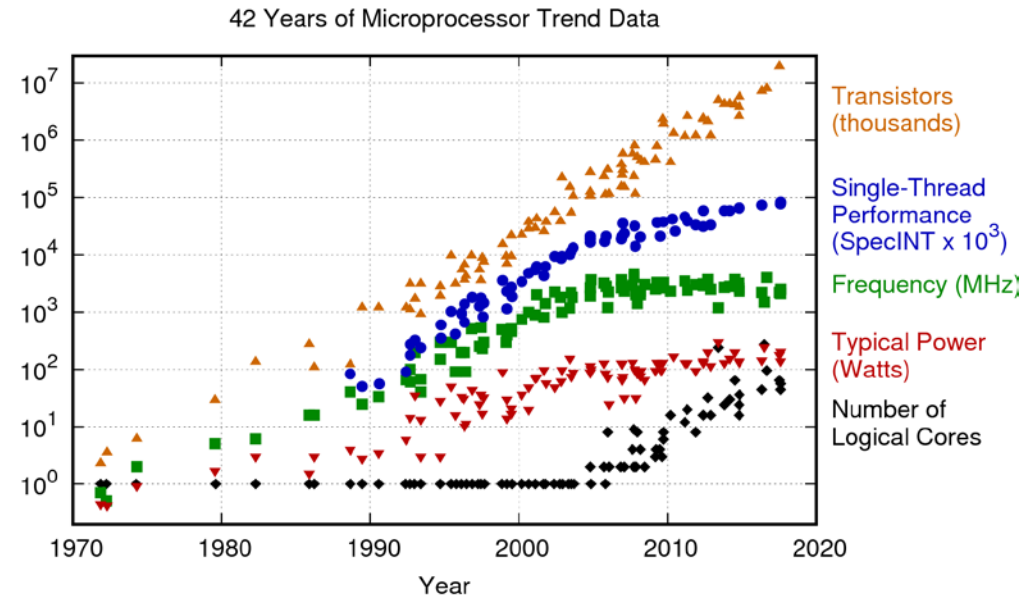
Testing

System

Hardware

Software

Increasing Semiconductor Complexity



Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2017 by K. Rupp

Different Tests

- Component
- Assembly
- Function

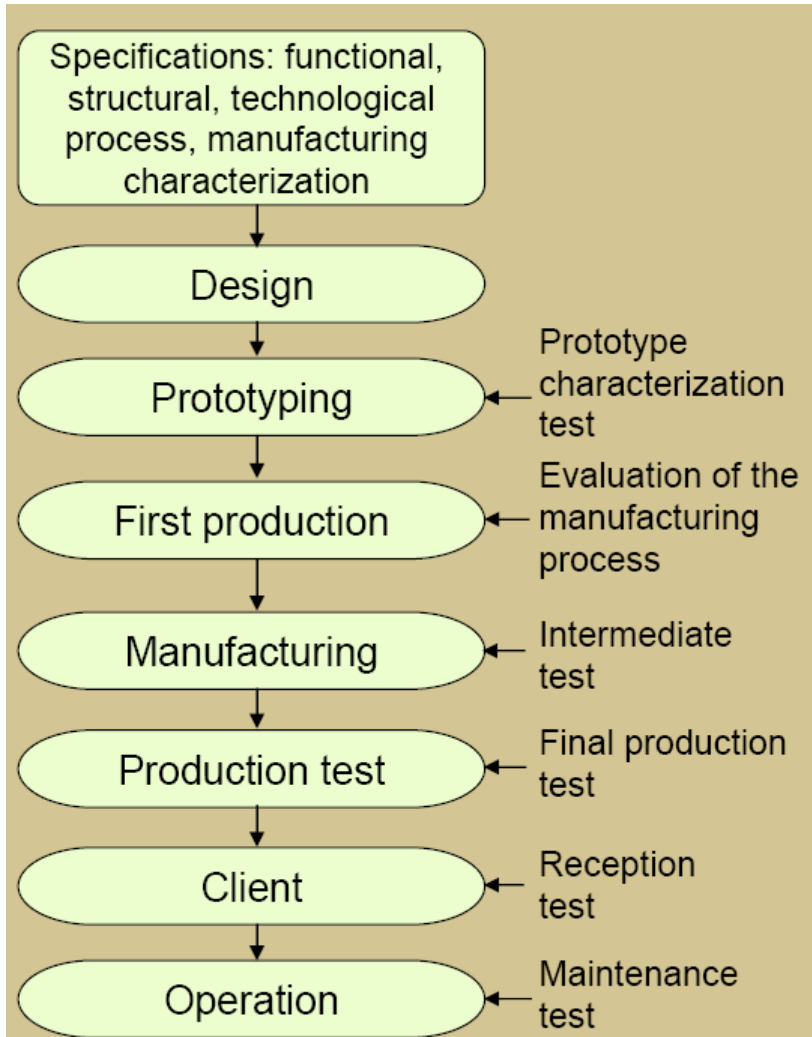
Different Stages

- Development
- Production
- During Use

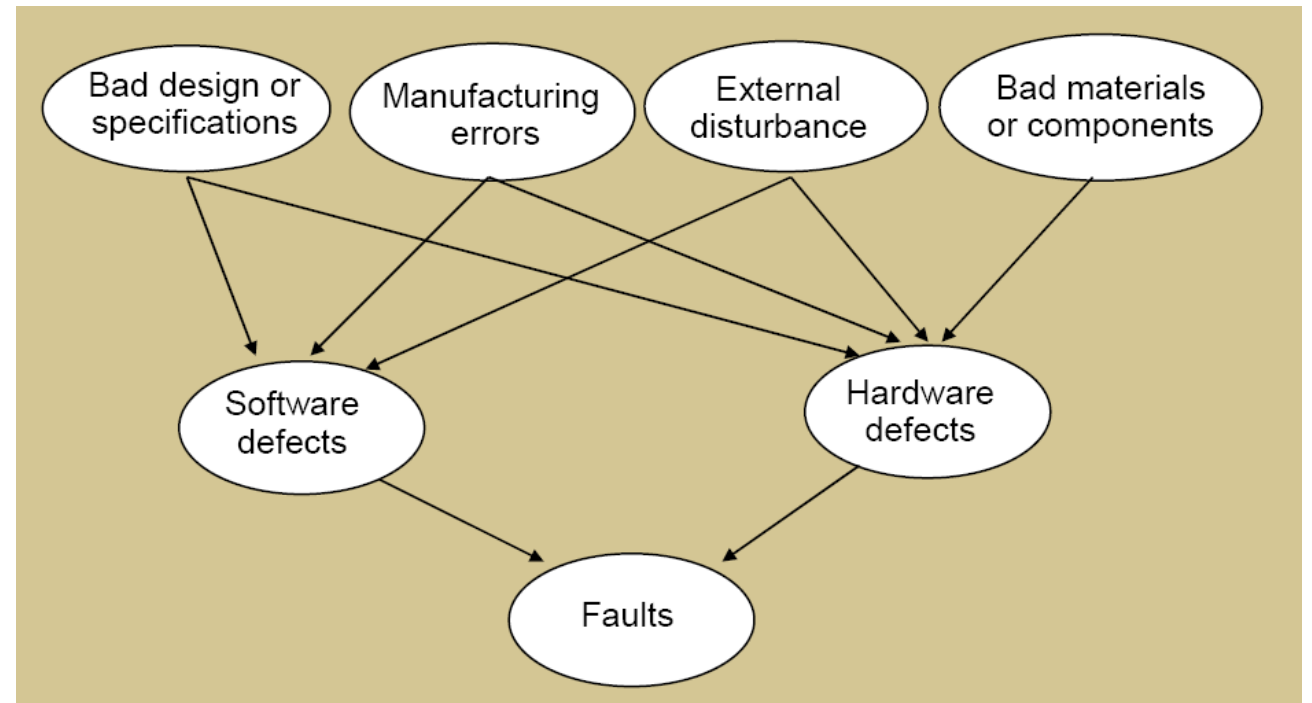
Increasing System Complexity

E.g. More than 100 uCs are used in a car. Whereas the most of them communicate with each other.

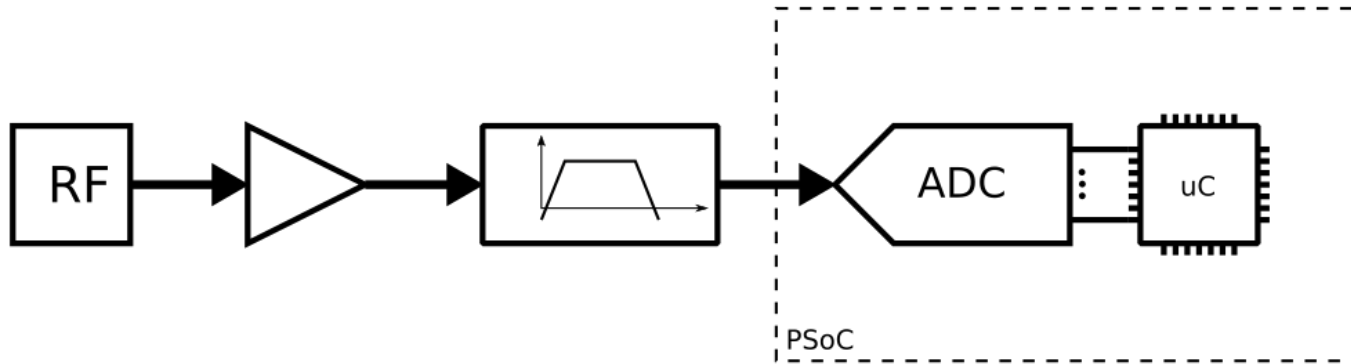
Product Life Cycle



How Faults Occur



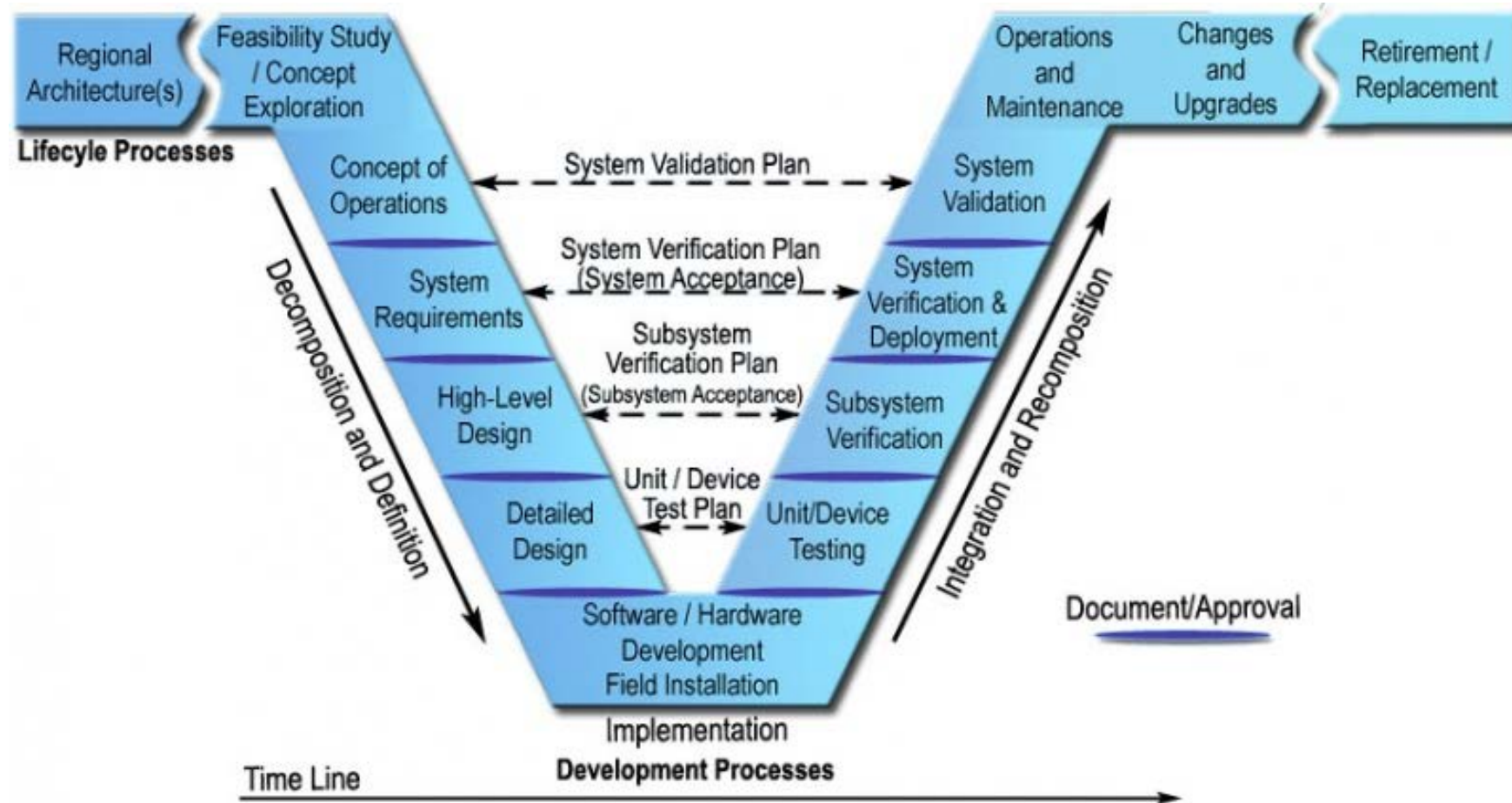
From: "Test and Design for Testability of Analog and Mixed-Signal Circuits ", José Machado da Silva



Which tests have to be done and where?

- Hardware
- Software
- System





Reference: (Caltrans and USDOT. 2005. Systems Engineering Guidebook for Intelligent Transportation Systems (ITS), version 1.1. Sacramento, CA, USA: California Department of Transportation (Caltrans) Division of Research & Innovation/U.S. Department of Transportation)

Agenda

- PCB – Testing
- Functional Circuit Test
- Software Test
- Build In Self Test (BIST)

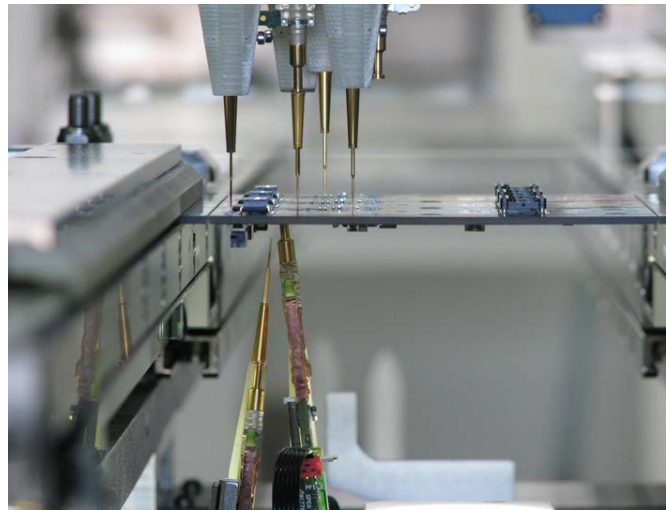
Multiple PCB Test Strategies

Different Test

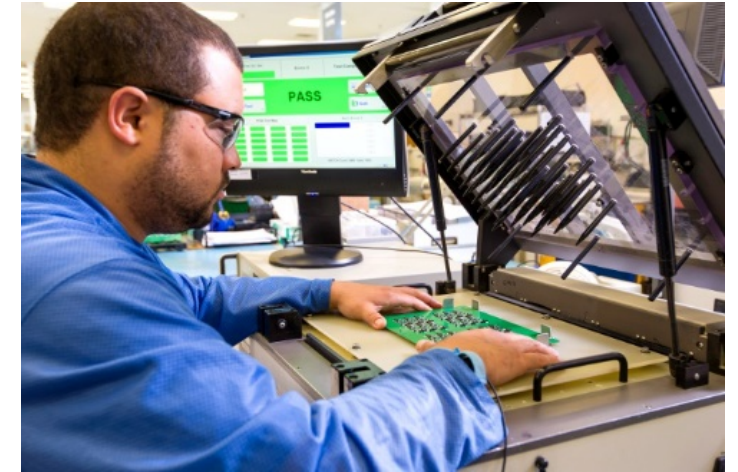
- In-Circuit Test (ICT)
- Fixtureless In-Circuit Test (FICT) / Flying Probe Test
- Functional Circuit Test
- Boundary Scan Testing
- Design for Manufacturing (DFM)
- Design for Assembly (DFA)
- Design for Test (DFT)
- Design for Supply Chain (DSC)
- Bypassing PCB Testing



Automatic optical inspection



Flying Probe Test



Fixtureless In-Circuit Test

What is being tested

- Lamination
- Copper Plating
- Solderability
- Hole wall quality

Common PCB Defects

DEFECT	RATE	TYPE	SOLDER-RELATED
Open	25%	Structural	Yes
Insufficient	18%	Structural	Yes
Short	13%	Structural	Yes
Missing electrical component	12%	Structural	No
Misaligned	8%	Structural	Yes
Defective electrical component	8%	Electrical	No
Wrong component	5%	Electrical	No
Excess solder	3%	Structural	Yes
Missing non-electrical component	2%	Structural	Yes
Wrong orientation	2%	Electrical	No
Defective non-electrical component	2%	Structural	No

Reference:

[“ Inspection and testing methods for PCBs: An overview”, Caltronics, Whitepaper #401](#)

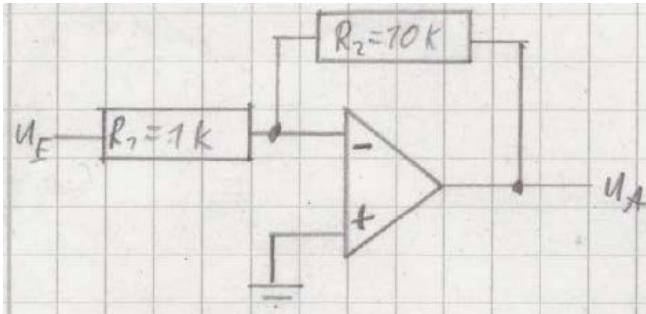
Circuit Defects and Inspection Methods

DEFECT TYPE	AXI	AOI	ICT
Soldering defects:			
Open circuits	Yes	Yes	Yes
Solder bridges	Yes	Yes	Yes
Solder shorts	Yes	Yes	Yes
Insufficient solder	Yes	Yes (not heel)	No
Solder void	Yes	No	No
Excess solder	Yes	Yes	No
Solder quality	Yes	No	No
Component defects:			
Lifted lead	Yes	Yes	Yes
Missing component	Yes	Yes	Yes
Misaligned/misplaced component	Yes	Yes	Yes
Incorrect component value	No	No	Yes
Faulty component	No	No	Yes
BGA and CSP defects:			
BGA shorts	Yes	No	Yes
BGA open circuit connections	Yes	No	Yes

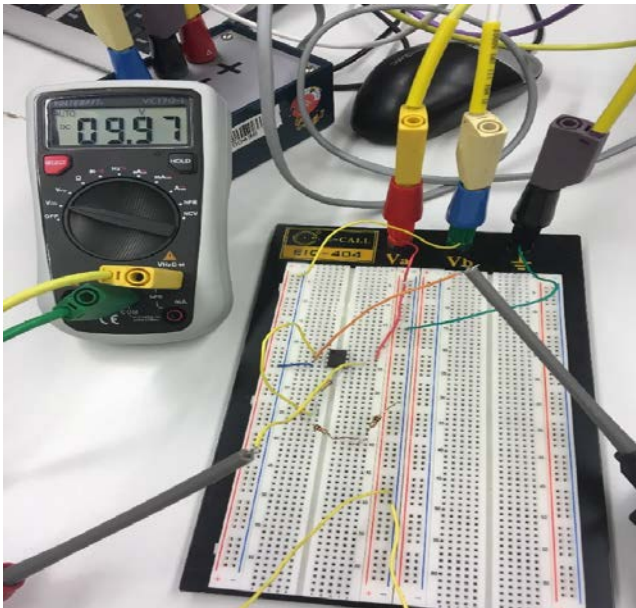
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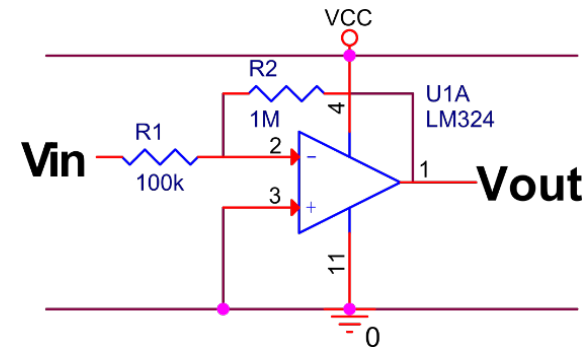
Functional Circuit Diagramm



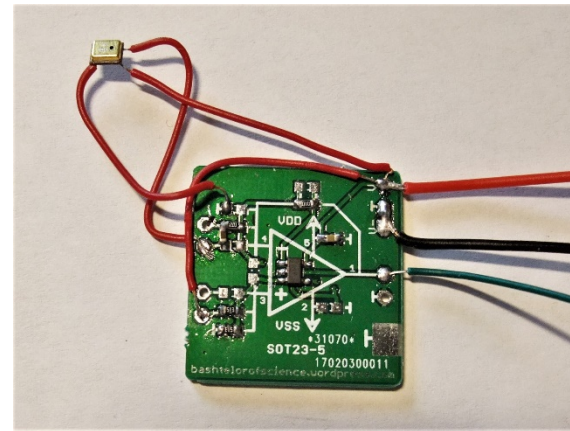
Bread Board Realisation



Schematic



PCB - Realization



Test Equipment

- Schematic and PCB Layout
- Pen and Calculator
- Volt/Current – Meter
- Power Supply
- Function Generator
- Oscilloscope
- Bread Board
- Resistors / LEDs / Switches

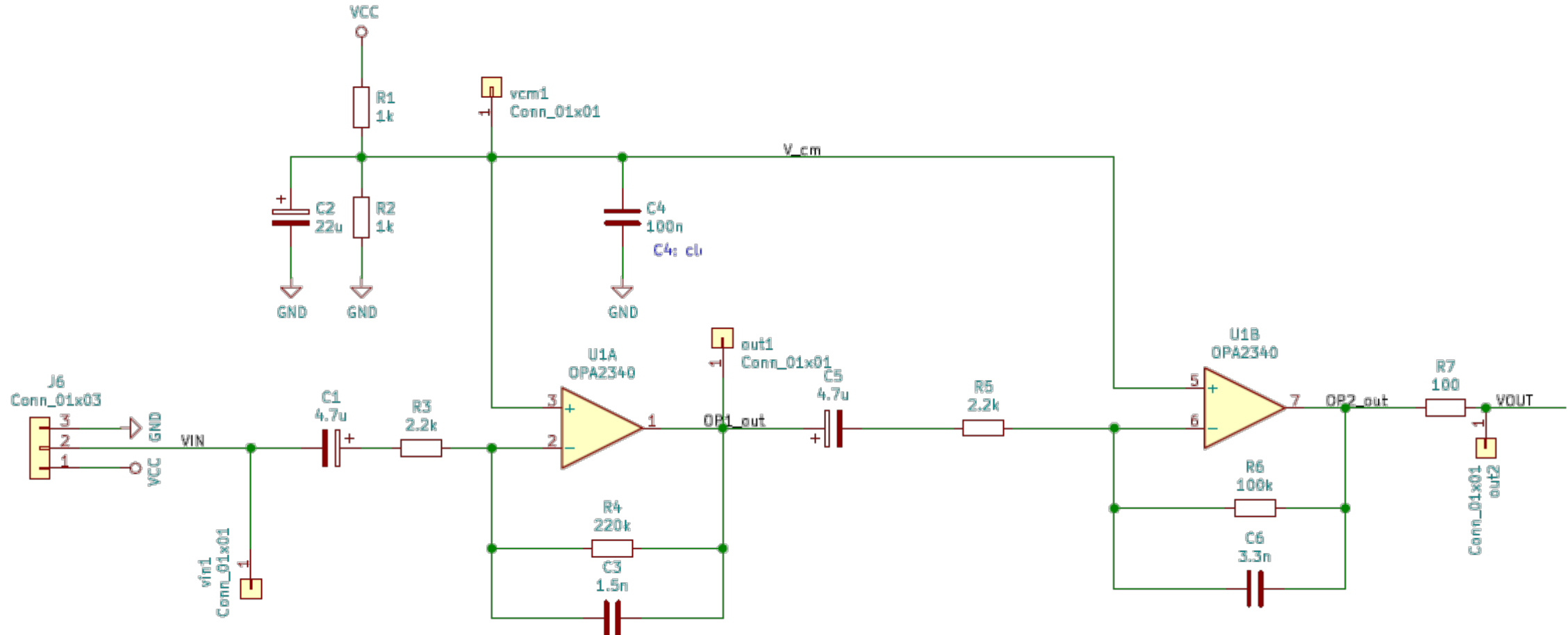
Excercise

What should be tested ?

How should it be tested ?

In which order should the tests be performed?

?



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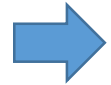
Questions by forehand

- What do you understand by testing?
- How do you perform software testing?

Definitions of Testing

- “Testing is the process of demonstrating that errors are not present.”
- “The purpose of testing is to show that a program performs its intended functions correctly.”
- “Testing is the process of establishing confidence that a program does what it is supposed to do.”

Adding a value to a program through testing means raising the quality and reliability of a program.
Raising the reliability of a program means finding and removing errors.



Testing is the process of executing a program with the intent of finding errors.

Common Misunderstanding

Goal is to find a test to find an error!
Successful / unsuccessful



You develop tests to find an error and not error freeness.
A successful test has shown an error.

Keep in Mind

- The concept of a program without errors is basically unrealistic.
- In general, it is impractical, often impossible , to find all the errors in a program.

During all development phases

- Specification
- Subroutines
- Libraries
- After Software Integration
- After Hardware – Software Integration

Machine Tests

Black Box

- Behavior and structure are unknown
- Test data solely derived from the specification
- Data-driven test
- Exhaustive input testing

White Box

- Internal structure is known
- Test data derived from the programs logic
- Logic driven test
- Exhaustive path testing

Derive Test Cases and Test Data

Human Tests

Inspection

- Reading statement by statement
- Group of programmers

Walkthrough

- Paper test cases
- Mentally executing the program
- Group of programmers

List of Errors

Team of four people

One moderator (this is NOT the author)

The author

Programmer

Test specialist

Duties of the moderator

- Distributing materials for, and scheduling, the inspection session
- Leading the session
- Recording all errors found
- Ensuring that the errors are subsequently corrected



Hints for
Your
Project

Aim

To find errors, not
correcting them

During the session, two activities occur

1. The programmer narrates, statement by statement, the logic of the program. During the discourse, other participants should raise questions, and they should be pursued to determine whether errors exist. It is likely that the programmer rather than the other team members will find many of the errors found during this narration. In other words, the simple act of reading aloud a program to an audience seems to be a remarkably effective error-detection technique.

2. The program is analyzed with respect to a checklist of historically common programming errors

Include Test possibilities in your Hardware / PCB Design

- Debug LEDs
- LCD Matrix
- UART Connection to a terminal program

- Push Buttons
- Switches

- Test Points

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BIST

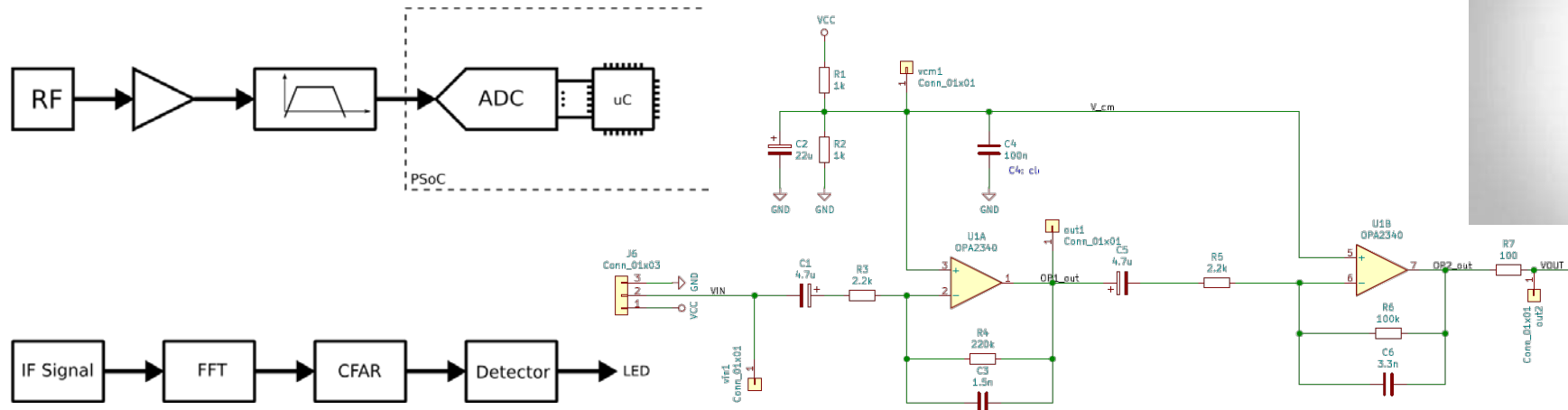
- Implemented in Hardware and Software to test the System, or special Hardware parts
- Performed during Power Up
- Performed frequently or occasionally during operation

Exercise

If you look at our system (Hardware, Software, Analog, Digital, Signal Processing, Flow Control, PCB, PSoC, PCB, Housing, etc.)

Answer the following question

- What would you like to test with a BIST?
 - What BIST can you implement?
- For each question find three points.



My first program in C

```
#include <stdio.h>
int main()
{
    printf("Hello World!");
    return 0;
}
```

Implement

- Test Points to your PCB Design
- Implement three debug LEDs with different color
- Implement a push button
- Implement a test signal input from and PSoC5LP pin to the Amplifier first and second stage.
- Discuss the implemented test points with the whole team.



Literature

- *“The Art of Software Testing, Second Edition”*, Glendford J. Myers, John Wiley & Sons, 2004
- *“Inspection and testing methods for PCBs: An overview”*, Caltronics, White Paper #401
- *“Survey of Electronic Hardware Testing Types, ATE Evolution & Case Studies”*, Manjula C and Dr. D. Jayadevappa, International Journal of Engineering Research & Technology (IJERT), 2017
- *“Test and Design for Testability of Analog and Mixed-Signal Circuits”*, José Machado da Silva, U.Porto – Faculdade de Engenharia INESC Porto, 2010